INCH-POUND
MIL-DTL-23053/18B
17 June 1996
SUPERSEDING
MIL-I-23053/18A
01 May 1986

DETAIL SPECIFICATION SHEET

INSULATION SLEEVING, ELECTRICAL, HEAT SHRINKABLE, MODIFIED FLUOROPOLYMER, CROSSLINKED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the sleeving described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-DTL-23053.

REQUIREMENTS:

Polymer type: The base polymer used in formulating this sleeving shall be a copolymer or terpolymer of polyvinylidene fluoride.

Continuous operating temperature range: Class 1: -55°C (-67°F) to +175°C (347°F). Classes 2 and 3: -55°C (-67°F) to +150°C (+302°F).

Color: The sleeving shall be supplied in colors and clear. Colors shall conform to the requirements of Class 1 of MIL-STD-104.

Class: The sleeving shall be furnished in the following classes, as specified (see 6.2a):

Class 1 - Semi-rigid. (Secant Modulus over 100,000 psi (690(Mpa).

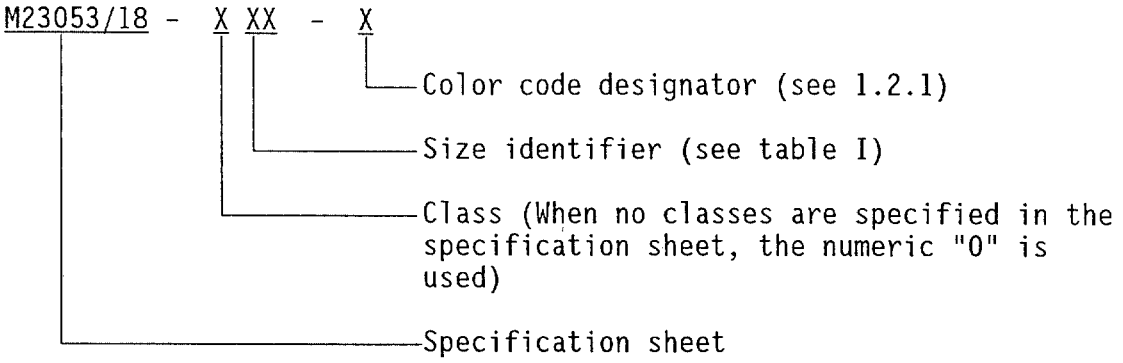
Class 2 - Flexible. (Secant modulus over 25,000 to 100,000 psi (172 to 690 Mpa)

Class 3 - Very flexible. (Secant modulus to 25,000 psi (172 Mpa) Maximum.)

Longitudinal Change: +5, -10 percent; overexpanded sleeving: -15 percent.

Military part number: The Military part number shall consist of the basic number of this specification sheet and dash numbers as follows:

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Example: Class 1, 0.250 inch (6.35 mm) as supplied ID sleeving shall be identified as M23053/18-106-C.

TABLE I. Construction details, inches (mm). 1/

Military part number	As supplied ID minimum	After unrestricted shrinkage	
		ID maximum	Wall thickness 2/
Class 1			
M23053/18-101-*	.046(1.17)	.023(.58)	.010 ± .002(.254 ± .051)
M23053/18-102-*	.063(1.60)	.031(.79)	.010 ± .002(.254 ± .051)
M23053/18-103-*	.093(2.36)	.046(1.17)	.010 ± .002(.305 ± .051)
M23053/18-104-*	.125(3.18)	.062(1.58)	.010 ± .002(.305 ± .051)
M23053/18-105-*	.187(4.75)	.093(2.36)	.010 ± .002(.305 ± .051)
M23053/18-106-*	.250(6.35)	.125(3.18)	.012 ± .003(.356 ± .076)
M23053/18-107-*	.375(9.53)	.187(4.75)	.012 ± .003(.356 ± .076)
M23053/18-108-*	.500(12.7)	.250(6.35)	.012 ± .003(.356 ± .076)
M23053/18-109-*	.750(19.1)	.375(9.53)	.017 ± .003(.432 ± .076)
M23053/18-110-*	1.000(25.4)	.500(12.7)	.019 ± .003(.483 ± .076)
M23053/18-111-*	1.500(38.1)	.750(19.1)	.020 ± .003(.508 ± .076)
M23053/18-112-*	2.000(50.8)	1.000(25.4)	.020 ± .003(.508 ± .076)
Class 2			
M23053/18-201-*	.046(1.17)	.023(.58)	.010 ± .002(.254 ± .051)
M23053/18-202-*	.063(1.60)	.031(.79)	.010 ± .002(.254 ± .051)
M23053/18-203-*	.093(2.36)	.046(1.17)	.010 ± .002(.305 ± .051)
M23053/18-204-*	.125(3.18)	.062(1.58)	.010 ± .002(.305 ± .051)
M23053/18-205-*	.187(4.75)	.093(2.36)	.010 ± .002(.305 ± .051)
M23053/18-206-*	.250(6.35)	.125(3.18)	.012 ± .003(.356 ± .076)
M23053/18-207-*	.375(9.53)	.187(4.75)	.012 ± .003(.356 ± .076)
M23053/18-208-*	.500(12.7)	.250(6.35)	.012 ± .003(.356 ± .076)
M23053/18-209-*	.750(19.1)	.375(9.53)	.017 ± .003(.432 ± .076)
M23053/18-210-*	1.000(25.4)	.500(12.7)	.019 ± .003(.483 ± .076)

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TABLE I. Construction details, inches (mm). 1/ - Continued

Military part number	As supplied ID minimum	After unrestricted shrinkage	
		ID maximum	Wall thickness 2/
OVEREXPANDED SLEEVING			
M23053/18-211-*	.093(2.36)	.031(.79)	.017 ± .003(.432 ± .076)
M23053/18-212-*	.187(4.75)	.062(1.58)	.020 ± .003(.508 ± .076)
M23053/18-213-*	.375(9.53)	.125(3.18)	.020 ± .003(.508 ± .076)
M23053/18-214-*	.750(19.1)	.250(6.35)	.025 ± .003(.635 ± .076)
M23053/18-215-*	1.000(25.4)	.333(8.46)	.030 ± .005(.762 ± .127)
Class 3			
M23053/18-301-*	.046(1.17)	.023(.58)	.010 ± .002(.254 ± .051)
M23053/18-302-*	.063(1.60)	.031(.79)	.010 ± .002(.254 ± .051)
M23053/18-303-*	.093(2.36)	.046(1.17)	.010 ± .002(.305 ± .051)
M23053/18-304-*	.125(3.18)	.062(1.58)	.010 ± .002(.305 ± .051)
M23053/18-305-*	.187(4.75)	.093(2.36)	.010 ± .002(.305 ± .051)
M23053/18-306-*	.250(6.35)	.125(3.18)	.012 ± .003(.356 ± .076)
M23053/18-307-*	.375(9.53)	.187(4.75)	.012 ± .003(.356 ± .076)
M23053/18-308-*	.500(12.7)	.250(6.35)	.012 ± .003(.356 ± .076)
M23053/18-309-*	.750(19.1)	.375(9.53)	.017 ± .003(.432 ± .076)
M23053/18-310-*	1.000(25.4)	.500(12.7)	.019 ± .003(.483 ± .076)

1/ Diameter limits for object to be enclosed shall be as recommended in technical data.

2/ Wall thickness values are less when shrinkage is restricted.

3/ The asterisk in the part number shall be replaced by color code designations (see 1.2.1).

Unrestricted shrinkage: Test method 4.6.5: 200° ± 2°C (392° ± 4°F) for 3 minutes.

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TABLE II. Physical properties. 1/

Characteristic	Requirement	Test procedure and conditions
As supplied:		
ID, minimum	Table 1	4.6.3
Heat shock	No cracks, flowing or dripping	4.6.8 275° ± 3°C (527° ± 6°F)
Secant modulus, psi (MPa)		
Class 1	100,000(690) min	4.6.12.1, 2 percent strain, ASTM D882
Class 2	25,000(172.4) to 100,000 (690)	
Class 3	25,000(172.4) max	
Restricted shrinkage	No cracks	4.6.6.1.1 200° ± 2°C (392° ± 4°F)
Voltage withstand	Pass	4.6.6.2
Clarity stability	Pass	4.6.16 200° ± 1°C (392° ± 2°F), 24 hours
Concentricity, min.	70% (50% for over expanded sizes)	4.6.3.3
Melting Point, °C (°F), min.		ASTM D4951 2/
Class 1	140 (284)	
Class 2	140 (284)	
Class 3	90 (194)	
After unrestricted shrinkage:		
ID, maximum	Table 1	4.6.3
Wall thickness	Table 1	4.6.3

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TABLE II. Physical properties. 1/ - Continued

Characteristic	Requirement	Test Procedure and conditions
Tensile strength, psi (MPa), minimum Class 1 Class 2 Class 3	 3500 (24.3) 2500 1500 (9.4)	4.6.13 ASTM D638, 2 inches/min.
Ultimate elongation, percent, minimum	Class 1 200 Class 2-300 Class 3-200	4.6.13 ASTM D638, 2 inches/min.
Dielectric strength, volts/mil (Kv/mm), minimum	400(15.7)	4.6.2 ASTM D2671
Volume resistivity, ohm-cm, minimum	10^{11}	4.6.2 ASTM D876
Specific gravity, maximum	2.00	4.6.2 ASTM D792
Water absorption, percent, maximum	0.5	4.6.2 ASTM D570, 24 hrs. at 23°C
Corrosion	No corrosion	4.6.10.1 and 4.6.10.2 160° ± 2°C (320° ± 4°F), 16 hours
Low temperature flexibility	No cracking	4.6.7.1 -55° ± 1°C (-67° ± 2°F) 3/
Flammability	4/	4.6.14 Procedure C ASTM D2671
Heat resistance, property after: Ultimate elongation, percent, minimum	 Class 1-100 Class 2-250 Class 3-100	4.6.9 Class 1: 250° ± 3°C (482° ±5°F) for 168 hours. Classes 2 & 3: 225° ± 2°C(437° ± 4° F), for 168 hours
Fluid resistance, property after:		4.6.11

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TABLE II. Physical properties. 1/ - Continued

Characteristic	Requirement	Test procedure and conditions
Tensile strength, psi (MPa), minimum Class 1 & 2 Class 3	 2000 (13.9) 1000 (6.9)	
Elongation, percent, minimum	Class 1-100 Class 2-250 Class 3-100	
Dielectric strength, volt/mil (Kv/mm), minimum	400(15.7)	
Fungus resistance	Rating of 0	4.6.2 ASTM G21

1/ Unless otherwise specified, requirements, test procedures, and conditions are for all classes.

2/ Test procedure in accordance with ASTM D4591. Melting point calculated as specified in "Melting endotherm peak" of ASTM D3222.

3/ Low temperature flexibility: Specimens 0.187 and smaller shall be tested as whole sections after unrestricted shrinkage. Specimens larger than 0.187 inch (4.75mm) shall be a 0.250 inch (6.35mm) wide strip cut from the sleeving. All specimens shall be tested after unrestricted shrinkage. Mandrels shall be 10 times the specimen O.D. ± 10% for tubular specimens or 10 times the specimen thickness ± 10% for strip specimens.

4/ Record whether any flaming or glowing particles or flaming drops fall from the specimen. If more than 25 percent of the indicator flag is burned away or charred (brown scorching and soot shall be ignored) after all of the five applications of flame or if flaming or glowing particles or flaming drops fall from the specimen at any time and ignite the cotton (flameless charring of the cotton shall be ignored), the specimen shall be judged to have conveyed flame and shall be considered as failing. In addition, any specimen which continues to flame or glow from one flame application longer than one minute shall not be acceptable.

Storage life conditions: Supplier shall certify to storage at 18° to 35°C (65° to 95°F) for 2 years. Conformance to 3.5. See 3.5.2 for storage life extension.

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Intended use:

MIL-I-23053/18 sleevings are intended for use as wire markers, insulation and wire and component relief. Clear materials may be considered for use as replacements for non flame retarded clear sleevings.

Custodians:
Army - ER
Navy - AS
Air Force - 85

Review activities:
Army - ME, MI, MU
Navy - EC, MC
Air Force - 80, 99

Preparing activity:
Navy - AS

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